



Analyzing player reviews with natural language processing to identify ecogames for education and research

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ARTICLE INFO

Keywords:

Climate change
Environmental education
Video games
Steam
Large language models
Artificial intelligence

ABSTRACT

Ecogames – games that address environmental topics – are an innovative and engaging method for fostering environmental education, which is urgently needed to respond effectively to climate change. However, educators and researchers are challenged by the task of selecting and investigating appropriate games from the vast and ever-growing assortment available. The rapid pace and large volume of the video game market complicate the selection process for classroom activities or research, such as media comparison studies. This study introduces a novel methodology that employs natural language processing techniques to identify and preselect ecogames that feature climate change-related content to varying degrees. By analyzing over 186,000 player-generated reviews on the Steam gaming platform through dictionary-based analysis and climate change detection classification, the methodology effectively distinguishes ecogames from action/adventure games and demonstrates varying levels of climate change-related content across different game categories. It also highlights that not all ecogames feature discussion on environmental matters equally, underscoring the need for careful game selection. Additionally, this approach has uncovered new ecogame candidates among a random selection of action/adventure games. Relying solely on the lexical features of the texts investigated, this study's approach is quick and scalable in its implementation and represents a data-driven way of preselecting ecogames based on publicly available text data. Combining the text analyses with other game metrics such as playtime, game popularity and age rating, the study recognizes *Beecarbonize*, *Alba: A Wildlife Adventure* and *Gibbon: Beyond the Trees* as notable candidates for in-depth research and potential educational use in the classroom.

1. Introduction

1.1. Climate change and video games

Climate change is internationally perceived as a primary threat (Poushter & Huang, 2019) causing harm to oceanic ecosystems, compromising food and water security, and leading to human morbidity and mortality due to extreme heat events, while also damaging economies across various sectors (Calvin et al., 2023). Particularly affected are vulnerable communities that have contributed the least to anthropogenic climate change (Calvin et al., 2023). Both top-down policies and widespread bottom-up engagement are crucial in meeting emissions reduction goals and securing a livable and sustainable future for all (Hart & Feldman, 2016). Environmental education is pivotal in promoting pro-environmental attitudes and behaviors (Varela-Candamio et al., 2018). This, however, can be particularly challenging on the topic

of climate change due to its abstract, slow-moving and complex nature that often conflicts with our psychological limitations (Gifford, 2011). These limitations include the human tendency to deprioritize events that are distant in time and space, difficulties in understanding uncertainty and complexity, the use of denial and rationalization to avoid cognitive dissonance or threats to the status quo, and the low salience of gradual and imperceptible changes (Gifford, 2011; Griskevicius et al., 2012; Ross et al., 2016; van Vugt et al., 2014). Accordingly, novel avenues for fostering environmental education and engagement, such as game-based learning (Boncu et al., 2022; Douglas & Brauer, 2021; Ndulue & Orji, 2023), are being explored to address the challenge of climate change. Some proposed elements of video games like *visual representation* (Flood et al., 2018), *increased motivation through engaging experience* (Koivisto & Hamari, 2019), and *learning through active experimentation* (Plass et al., 2015) can explain their potential to go beyond pure entertainment outcomes and facilitate learning experiences by making the topic of

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<https://doi.org/10.1016/j.chbr.2025.100850>

Received 9 July 2025; Received in revised form 25 October 2025; Accepted 29 October 2025

Available online 1 November 2025

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climate change more tangible and interesting.

Recent meta-analytic evidence highlights that video games, particularly those employing persuasive mechanics can effectively induce attitudinal change towards the topics depicted in the video game, such as violence, social justice, health, and environmental issues (Kolek et al., 2023). As such, games and game design principles show promise in fostering pro-environmental attitudes and behaviors (Douglas & Brauer, 2021; Fernández Galeote et al., 2021, 2023; Fernández Galeote & Hamari, 2021; Janakiraman et al., 2021; Zhang et al., 2024). The field of environment-related game-based approaches is diverse, encompassing analog games, digital games and hybrid formats that blend elements of both (Fernández Galeote et al., 2021). These formats range from analog tabletop or role-playing games to hybrid role-plays using modeling software and digital video games, naturally varying in their use of game elements. While achievement- and immersion-oriented elements are common across all formats, social elements are less prevalent in digital games (Fernández Galeote et al., 2021). Another key distinction within environment-related game-based approaches lies between serious games – designed with specific educational or purposeful intent – and entertainment games. Although entertainment games can be repurposed for serious objectives, this requires a shift from their original intent (Djaouti et al., 2011). Entertainment video games have demonstrated promise as educational tools across a wide range of academic disciplines, particularly in foreign language learning and science education (Martínez et al., 2022). They also show potential for enhancing engagement with climate change topics (Fernández Galeote & Hamari, 2021). Fernández Galeote and Hamari (2021) specifically recommend moving beyond a game's intended purpose in favor of systematically examining individual video games. Similarly, Kolek et al. (2023) emphasize that while narrative video games – those that incorporate storytelling elements – are primarily perceived as sources of entertainment, they can also serve as powerful tools for information dissemination, shaping how players understand and interpret the topics depicted in-game. Even 360° nature footage videos that incorporate storytelling elements have been shown to be effective tools for environmental communication (Amrhein & Balaban, 2024). Beyond that, narrative video games function as interactive information systems, enabling players to engage with presented information, observe the consequences of their actions within the game world, and respond to those changes (Kolek et al., 2021). Kolek et al. (2023) further highlight that persuasive game mechanics (e.g., *perspective-taking*, *meaningful feedback*) can influence attitudes regardless of the game designers' intentions, underscoring the potential of entertainment games as tools for attitudinal and behavioral change.

Video games designed to raise environmental awareness and promote environmental education are categorized as *ecogames* (De Beke et al., 2024). Ecogames or alternatively also referred to as *green games*, *climate games* or *eco-critical games*, however represent a loosely defined category of video games that extends beyond the realm of purely serious games. This category encompasses a wide variety of video games questioning human-environment relations, simulating environmental challenges or reflecting present socio-environmental conditions (De Beke et al., 2024). Consequently, even markedly distinct games such as *Beecarbonize* (Beecarbonize, 2023), *The Wandering Village* (*The Wandering Village*, 2022; *Battlefield 2042*; *Battlefield 2042*, 2021) can be pooled together under this term (De Beke et al., 2024).

With video games growing from niche interests to culturally pervasive artifacts reaching a global audience of 3.4 billion players in 2024 (Buijsman, 2024), ecogames hold special potential as a medium for environmental education and climate change communication. The combination of a vast outreach and the motivating and engaging elements present in video games (Koivisto & Hamari, 2019) can create an ideal platform for promoting pro-environmental attitudes and behaviors, including among younger generations (Douglas & Brauer, 2021). However, educators aiming to include ecogames in classrooms to, for instance, promote discussions on environmental issues face the daunting

task of selecting suitable titles from the multitude of games released each year on various platforms. In 2024 alone, Steam, the largest PC game distribution platform, released 18,869 games, a staggering increase of 1100 % over one decade as compared to 1,713 games in 2014 (SteamDB Releases, n.d.). Even researchers planning to study the effects of ecogames, for instance, in media comparison studies (Mayer, 2019), might struggle to identify the most suitable games due to the sheer number of new video games published each year. In a similar vein, the rapid pace of game releases complicates video game research in general, as new games often emerge before studies can publish their findings and communicate them to relevant stakeholders, such as educators. In other words, the vast number of available games complicates the process of selecting specific ecogames particularly for non-gaming experts.

1.2. Related work

Importantly, however, video gamers frequently leave digital footprints, in the form of player-generated reviews, notably on platforms like Steam, which boasts over 125 million reviews (Steam250, n.d.). These reviews offer rich insights into the players' feelings, attitudes towards the games and their effects (Martens & Johann, 2017; Ninaus et al., 2025; Pagano & Maalej, 2013; Phillips et al., 2021; Tong, 2021; Zhu & Fang, 2015) and thus may assist in the preselection process of notable ecogames.

Lin et al. (2019) conducted an extensive empirical analysis of 10,954,956 Steam reviews across 6,224 games, uncovering several key insights into the structure and content of player-generated reviews. They highlighted the importance of both positive and especially negative reviews for developers, noting that negative reviews are often posted after only half the playtime of positive reviews. Similarly, Guzsvinecz and Szűcs (2023), in their analysis of 35,983,481 Steam reviews, confirmed that negative reviews are typically written earlier than positive ones across various game genres. They also found that negative reviews tend to be longer, further underscoring their significance for developers. Additionally, their sentiment analysis revealed that the emotional tone of reviews varies across different top-level game genres on Steam.

Phillips et al. (2021) explored the therapeutic potential of commercial video games by qualitatively analyzing player-written reviews of the top 50 highest-rated games on Steam. They discovered that players frequently reported mental health benefits, such as using games to relax, manage mental health challenges, and strengthen relationships with friends and family. Based on these reviews, the researchers identified a shortlist of seven games with therapeutic potential for further study. Similarly, Kohlburn et al. (2023) qualitatively analyzed 400 player-written reviews of four LGBTQ + inclusive games on Steam. They identified 22 recurring themes and found a generally positive (though not unanimous) perception of LGBTQ + representation in these games.

A broader study by Busurkina et al. (2020) examined 613,195 reviews of 335 games using Structural Topic Modeling. This analysis identified seven overarching clusters of topics (i.e., Achievement, Narrative, Social Interaction, Social influence, Visual/Value, Accessories, General experience). Xing et al. (2025) focused on 63,818 reviews of *Stardew Valley*, employing sentiment analysis, Latent Dirichlet Allocation (LDA), and GPT-assisted mechanism tagging. They concluded that players predominantly use the game as a form of emotional relief, with escapism significantly predicting playtime. Petrosino et al. (2022) investigated the evolution of video game reviews from 2018 to 2020, particularly in the context of the COVID-19 pandemic. They observed an increase in gaming activity during the pandemic, especially for games tagged as "Casual" and "Sport," suggesting an influx of new players. Additionally, reviews during this period became shorter and less helpful, though not more negative compared to 2018 and 2019.

The classification of sentiment in Steam reviews has also been a focus of research. Vigiato et al. (2022) identified challenges in sentiment classification, such as the use of sarcasm and contrastive conjunctions (e.g., "but," "although"). However, by 2024 Vigiato and Bezemer reported

that these issues had largely been resolved with the advent of large language models (LLMs).

1.3. Our approach

This paper aims to harness these player-written reviews and introduce an innovative, scalable, and data-driven methodology for pre-selecting promising ecogames that feature climate change-related content based on natural language processing (NLP) techniques. NLP is a branch of artificial intelligence used for a fast and automatic extraction of information from large amounts of unstructured text (Jonnalagadda et al., 2015; Roldan-Baluis et al., 2022). In this study, a LLM-extended dictionary-based approach and climate change detection classification using the domain-adapted ClimateBERT (Webersinke et al., 2022) model is employed to assess the level of climate change-related content of player-generated reviews of different games. The use of NLP techniques allows for the analysis of vast amounts of qualitative data that would be infeasible to process manually, enabling a more comprehensive and representative understanding of the population of games and their reviews. By combining the NLP-derived insights with other game attributes available on Steam, such as the popularity, average playtime and age rating, a further refinement of game selection to a few notable ecogames is enabled, even for non-experts in the field of gaming. This study focuses specifically on commercially available video games on the gaming platform Steam. Following the recommendations of Fernández Galeote and Hamari (2021), this study prioritizes a systematic examination of individual video games over an analysis based solely on the designers' intended purpose. However, focusing exclusively on games available on Steam inherently excludes many games that are not commercially available, such as those originating from game jams, most games developed by small independent creators, and certain serious games.

The primary objective of this study is to introduce a scalable NLP methodology for identifying and preselecting content-relevant ecogames that moves beyond existing game lists and eliminates reliance on game descriptions, opting instead for first-hand reports from players in the form of game reviews. Structuring the game selection process for studies, by basing it on large bodies of qualitative data rather than on researchers' personal preferences may improve the overall quality of future research findings (Tyack et al., 2018). Additionally, leveraging player-generated reviews and NLP could enable researchers to keep pace with the rapid developments of the gaming industry and dispense with potentially outdated game lists. Selecting appropriate and content-relevant ecogames appears to be vital for educational and research purposes, given the variety of games encompassed by the term 'ecogames' (De Beke et al., 2024), which address climate change-related content to varying degrees. It is important to note that this study does not evaluate the games' ability to evoke discussions on climate change or related topics in classroom settings. Instead, we treat player-written reviews containing climate change-related content as unsolicited instances of such discussions. Accordingly, we hypothesize that games with higher levels of climate change-related content in their reviews are more promising for educational purposes, particularly in the context of reflection and debriefing sessions.

Debriefing is widely regarded as one of the most critical activities in any lesson that incorporates games as learning tools (Squire, 2002). However, the omission of debriefing activities – an issue that occurs frequently with educational games – can hinder the transfer of knowledge and insights from the game to real-life contexts (Nicholson, 2012). Team-based debriefings, which often involve first discussing game-related questions collaboratively and then writing individual reflections, have been shown to produce better outcomes than self-directed debriefing (Bilgin et al., 2015). Games that naturally generate higher levels of discussion around climate change and related topics in player-written reviews may therefore hold particular value in classroom settings, where team debriefings can foster deeper

engagement and understanding.

1.4. Aims and research questions

On a broader scale, our study aims to contribute to the field of Human-Computer Interaction (HCI) (Preece et al., 1994) by demonstrating how Natural Language Processing (NLP) techniques can be used to analyze player-written reviews as a rich source of behavioral insights. Player-written reviews provide valuable information across multiple dimensions: (1) behavioral metrics, such as playtime (Xing et al., 2025); (2) sentiment, reflected in the valence of the review (Viggiano et al., 2022); and (3) textual information, freely expressed by users without external constraints. These reviews represent digital traces voluntarily left by users, with their textual components providing a valuable resource for analyzing key human variables that shape behavior. By examining these reviews, researchers can infer latent variables that are central to understanding human behavior in general, including mental health (Phillips et al., 2021), LGBTQ+ (Kohlburn et al., 2023) issues, and, in our specific case, engagement with climate change-related content. Our research thus also aims to illuminate how individuals generally interact with video games and digital platforms, such as Steam, to express their thoughts and experiences. Importantly, it addresses a significant social issue – climate change – by exploring how it is represented and discussed in player-written reviews of various ecogames, underscoring the connection between societal values, such as environmental protection, and gaming. By analyzing playtime and sentiment data, the study investigates a link between user engagement and feedback, which are essential also in other research areas like evaluating the adoption and impact of technologies (Hussain et al., 2019). Lastly, by applying NLP as a methodological tool, this research aims to provide valuable insights into user behavior and experiences, demonstrating the potential of AI-driven methods, such as NLP, to analyze and interpret human-generated content in digital environments on a large scale.

Specifically, for the current study, over 186,000 player-written reviews of 45 ecogames and 45 action/adventure games are compared to contrast levels of climate change-related content between these categories of games (for the selection procedure of games see Methods subsection 2.1.). This comparison serves two purposes: first, to demonstrate the capability of NLP techniques to differentiate between these categories, and second, to validate the ecogame category as a meaningful classification. It is important to note that "ecogame" is not a standalone genre, and some of the randomly selected action/adventure games may also be considered ecogames. However, this overlap does not pose a problem for the context of this study, as the goal is to test the validity of the ecogame category against an established genre. If the differentiation between ecogames and action/adventure games holds, even with some overlap, it strengthens the credibility and robustness of the results. Action/adventure games were chosen as the control group because these tags are the second and third most popular on Steam, whereas the most popular tag, "Indie," does not represent a formal genre.

Furthermore, the variability of climate change-related content among ecogames is considered, with the expectation that games with a central focus on climate change and related topics, will contrast against those only tangentially addressing these topics. Consequently, we expect greater variability in the levels of climate change-related content among ecogame reviews compared to action/adventure game reviews. For action/adventure games, we anticipate uniformly low levels of climate change-related content. In contrast, the loose criteria defining the ecogame category are likely to result in a broader range of focus on climate change, leading to greater variability in the reviews. Finally, this paper explores our methodology's capability to uncover ecogames that have been previously unrecognized by the literature. Some of these games might have gone unnoticed due to the volume of the gaming market or they might be newer games that have not been incorporated into

scientific studies yet. The authors emphasize that the goal of this study is not to replace expert and human evaluation but to complement and inform it. Additionally, this work aims to lay the groundwork for future, more extensive investigations of the Steam platform.

Consequently, we hypothesize:

H1. The level of climate change-related content, as assessed by dictionary analysis and climate change detection classification, will be higher in the reviews of the ecogame category compared to the action/adventure game category.

H2. The level of climate change-related content, as assessed by dictionary analysis and climate change detection classification, will vary significantly more within the ecogame category compared to the action/adventure game category.

2. Methods

Analyses were conducted using Python version 3.12.8. All reviews quoted in this article were rephrased by ChatGPT-4-Turbo and subsequently reviewed manually to avoid copyright infringement and ensure

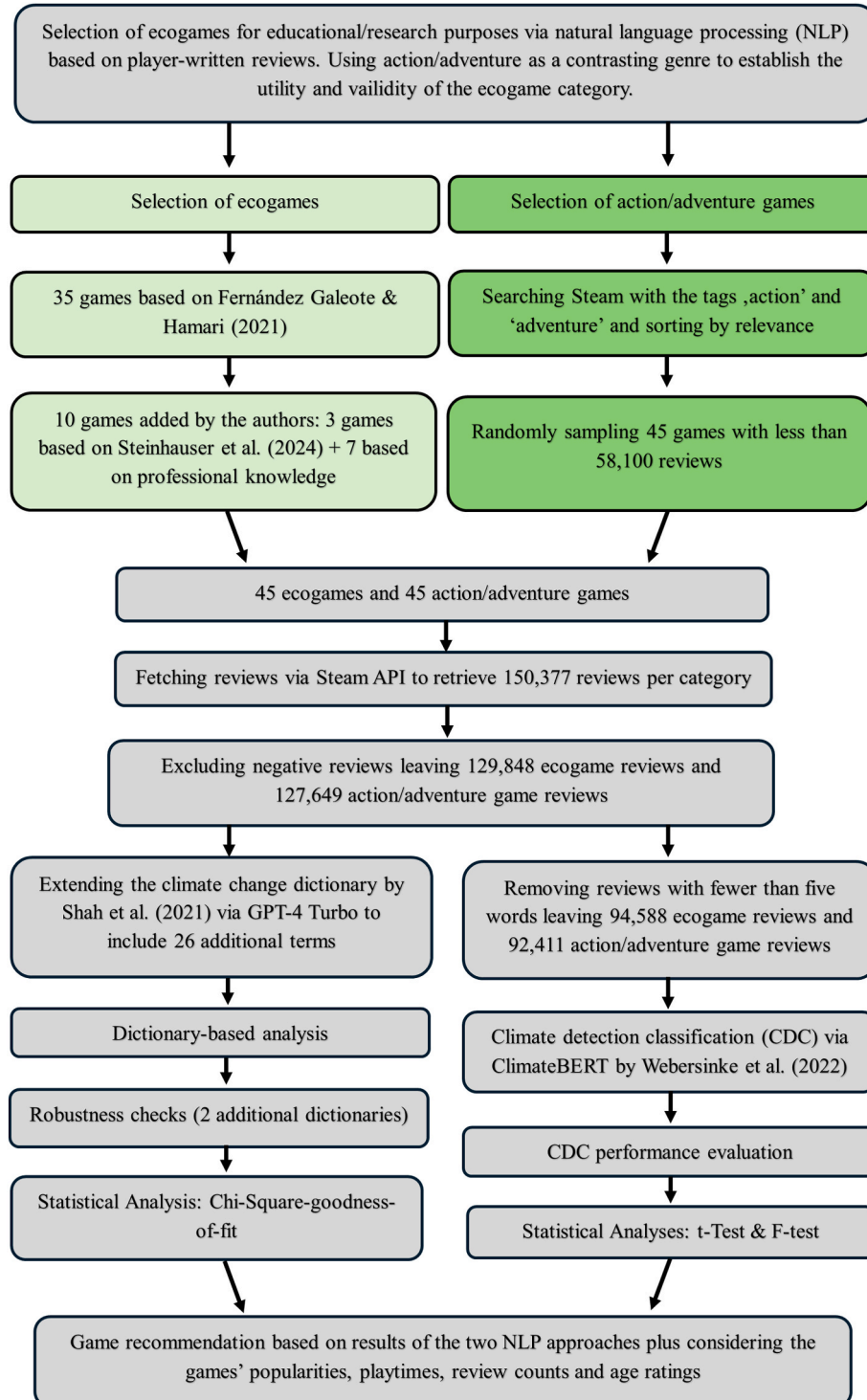


Fig. 1. Flowchart of the study procedure.

anonymity. The rephrasing utilized a GPT version provided by the authors' university, which operates on European Microsoft Azure servers, ensuring that prompts and data are processed within the EU and will not be used for training AI models. However, upon request, we can privately share the unmodified reviews with other researchers. The code used for this study, along with detailed instructions and comments to facilitate replication, is publicly available on OSF at https://osf.io/6hnpq/overview?view_only=13cbdb9ac829429187bb296f94241dab. The study procedure is illustrated in Fig. 1 through a flowchart.

2.1. Data collection

Data collection was carried out in November 2024, utilizing the Steam web-based API to retrieve player-generated reviews of predefined games. The selection of ecogames was based on a review article of Fernández Galeote and Hamari (2021) which identified 150 ecogames, both entertainment and serious games, across various platforms. Of these, 35 available on Steam were included in the analysis, along with 10 additional ecogames known to the authors through previous studies ($n = 3$) (Steinhauser et al., 2024) and professional knowledge ($n = 7$). The seven games included through professional knowledge are highlighted in gray in Table 1. These include *Alba: A Wildlife Adventure* (Alba:

Table 1

Games used in this study plus their review counts.

Games used in this study plus their review counts.

Ecogames		Action/Adventure Games	
Title	N	Title	N
Arma 3	58100	Assassins Creed Odyssey	53694
Mad Max	35238	Assassins Creed Valhalla	9899
Eco	7501	Far Cry 6	9356
Anno 2070	7249	Another Crab's Treasure	7391
Democracy 3	5394	SCARLET NEXUS	6222
Niche - a genetics survival game	4195	Freedom Planet	3523
Terra Nil	3635	Ember Knights	3439
Bear and Breakfast	3396	Forever Skies	2699
Redout: Enhanced Edition	3013	Senua's Saga: Hellblade II	2583
Alba: A Wildlife Adventure	2964	Unravel Two	2470
Anno 2205	2827	METAL GEAR SOLID 3: Snake Eater -	2350
The Descendant	2388	Rayman Origins	2316
Flower	1730	ASKA	2276
Civilization VI: Gathering Storm	1521	Moonstone Island	2237
Endling - Extinction is forever	1108	Mortal Sin	2229
Beecarbonize	1042	Goat Simulator 3	2152
Penguins Arena: Sedna's World	895	Chorus	2145
A New Beginning - Final Cut	820	The Entropy Centre	2003
Lumino city	811	Grand Theft Auto: San Andreas - The	1958
Forged Battalion	678	Planet of Lana	1946
Block'hood	639	Eiyuden Chronicle: Hundred Heroes	1862
UNDER the SAND - a road trip game	612	Compound	1822
Power & Revolution 2021 Edition	472	Wizard with a Gun	1782
Project Aura	435	Le Mans Ultimate	1770
The Eternal Castle [REMASTERED]	420	Dwarven Realms	1718
Greed Corp	400	Expeditions: A Mudrunner Game	1542
Dark Future: Blood Red States	389	PILGRIM	1511
Fate of the World: Tipping Point	380	Crash Bandicoot 4: It's About Time	1357
Plasticity	367	NARUTO X BORUTO Ultimate Ninja	1286
Gibbon: Beyond the Trees	320	Turnip Boy robs a bank	1251
The Aquatic Adventure of the Last Human	311	WRC Generations	1209
Cities: Skylines - Green Cities	243	Scars Above	1126
1000 days to escape	180	Observer: System Redux	1121
Koral	161	VLADiK BRUTAL	1077
Preserve	138	Star Trek: Resurgence	1059
Bionic Heart	85	Momodora: Moonlight Farewell	959
Your Future Self	82	Arizona Sunshine 2	929
The Sims 4: Eco Lifestyle	49	TerraTech Worlds	879
Tropico 5 Waterborne DLC	48	Lorelei and the Laser Eyes	748
President for a Day - Corruption	47	Prince of Persia The Lost Crown	696
Survival Tycoon	27	Kingdom Come: Deliverance - Band of	391
Ziro	25	Battle Shapers	382
Trash Time	24	The Talos Principle 2 - Road to Elysium	378
The Unclearness	18	Turok 3: Shadow of Oblivion Remastered	337
Bob & Kuura: Lost in Snowglobe	0	Welcome to Paradise	297

Note: Games are sorted in descending order based on the number of reviews available per game. The ten ecogames that were added by the authors are highlighted in gray.

A Wildlife Adventure, 2020), referenced in the study by Lu et al. (2023) and Plasticity (Plasticity, 2019), based on the study by Bekoum Essokolo and Robinot (2022). Three of the remaining five games were published after 2021 and were therefore not included by Fernández Galeote and Hamari (2021): Preserve (Preserve, 2024), Endling – Extinction is Forever (Endling - Extinction Is Forever, 2022) and Terra Nil (Terra, 2023). Finally, one game represents downloadable content (DLC): The Sims 4 – Eco Lifestyle (The Sims 4 Eco Lifestyle, 2020) and the last game, Anno 2205 (Anno 2205, 2015) was included as a logical continuation of Anno 2070; Anno 2070, 2011), which was part of the original review article by Fernández Galeote and Hamari (2021).

For these 45 ecogames, all reviews tagged as being in English were fetched from the platform, resulting in a total of 150,377 reviews. Similarly, 45 action/adventure games were randomly selected using the tags *action* and *adventure* and applying the ‘sort by relevance’ filter on Steam to match the number of ecogames. Sorting by relevance helped to avoid the inclusion of games with zero reviews, as the distribution of reviews on Steam is heavily skewed, with most games having 20 or fewer reviews (Lin et al., 2019; Zuo, 2018). To maintain consistency, no action/adventure games with more than 58,100 reviews – the maximum review count among the selected ecogames – were included. Subsequently action/adventure game reviews were randomly sampled to match the number of available reviews for ecogames, resulting in 150,377 action/adventure game reviews. The sampling was only done on reviews of the action/adventure game and was only necessary for the game with the highest review count, i.e., *Assassins Creed Odyssey* (Assassins Creed Odyssey, 2018) to preserve all the reviews for games with lower review counts. The games and respective review counts are listed in Table 1.

2.2. Data preprocessing

Writing a Steam review requires users to make a binary decision about recommending the game, which has previously been used as an indicator of review sentiment (Viggiato et al., 2022). For data preprocessing, ‘negative’ reviews were omitted from subsequent analyses, leaving 129,848 ecogame reviews and 127,649 action/adventure game reviews. The decision to omit negative reviews was based on the assumption that negative reviews might predominantly mention complaints about game design or bugs, (Lin et al., 2019), which is valuable for developers to improve their games (Moro et al., 2022), but has limited relevance for the current study. As such, negative reviews that mention environmental topics likely reflect dissatisfaction with gameplay, game design, or technical aspects, rather than providing insights into meaningful learning experiences. This rationale is exemplified by the following rephrased reviews:

‘This game places you on a server, to team up with others in a simulated ecosystem and build a society capable of stopping a planet-threatening asteroid. While this sounds new and engaging in theory, the user interface is confusing, and game controls are clunky. I really like the idea, but I became bored after spending hours on cutting trees without clear direction or goals, I think I’ll hold off until the game eventually fully releases.’ – review of *Eco* (Eco, 2018)

‘I like some of the game’s elements, but the way carbon emissions are handled seems off. You are in the 17th century, but if you’re the first to use coal and build a few power plants, suddenly the sea levels rise, and you get diplomatic penalties ... To me, climate change should come into play later in the game, and any penalties from international bodies should make sense, not just happen automatically even when no other civ is even aware of coal or industrialization.’ – review of *Civilization VI: Gathering Storm* (Civilization VI: Gathering Storm, 2019)

As part of the preprocessing before the dictionary-based analyses the reviews were converted to lowercase and special characters were

removed, using the regular expression pattern `[\w\s]`. Then multiple consecutive whitespaces were replaced by a single whitespace to facilitate word detections.

In preparation for the climate change detection classification, multiple new lines in the text were removed, as well as special characters using the regular expression pattern `[a-zA-Z0-9\.,!?:]`. This way only alphanumeric characters, whitespace and a defined set of punctuation marks were retained, with the goal of assisting the ClimateBERT model, which was not trained on social media data. Additionally, reviews consisting of fewer than five words were excluded, leaving 94,588 ecogame reviews and 92,411 action/adventure game reviews. This was done to enhance the classification’s quality because very short reviews typically lack substantial information and are unlikely to meaningfully feature climate change-related content. No further preprocessing was deemed necessary for the climate change detection classification because LLMs are adept at processing natural language (Schwitter, 2025). Finally, the ecogame *Bob & Kuura: Lost in Snowglobe* (Bob and Kuura: Lost in Snowglobe, 2019) was excluded from subsequent analyses, as no reviews were available on Steam.

2.3. Assessment of climate change-related content

To assess the level of climate change-related content per review, this study uses two established methodologies: 1) a dictionary-based approach and 2) climate change detection classification.

To maintain simple and consistent terminology with the original ClimateBERT study by Webersinke et al. (2022) and the climate change dictionary by Shah et al. (2021), we have chosen to use the label ‘climate change-related’ instead of the broader term ‘environmental-related’. For the purposes of this study, we do not want to delve into the semantic subtleties between these terms; rather, we emphasize that the label ‘climate change-related’ should be interpreted broadly to encompass environmental topics, even though climate change remains a focus of the paper. Both the dictionary-based approach and the climate change detection classification using ClimateBERT (Webersinke et al., 2022) extend beyond a narrower interpretation of the term ‘climate change’. This is why we use the term “climate change-related” – it suggests a primary focus on climate change while allowing for a broader scope that also encompasses related environmental and sustainability topics, that might have been overlooked otherwise. Additionally, the central role of climate change among environmental issues further informed this choice of terminology.

2.3.1. Dictionary-based approach

The aim of the dictionary analysis is to detect occurrences of pre-defined terms within a text corpus. This approach leverages an existing dictionary formulated with specific climate change-related terms, further expanded with the assistance of an LLM. The original dictionary was developed based on discussions by Reddit users and terms commonly used in the media, containing 26 terms (e.g., ‘climate change’, ‘global warming’, ‘co2 emissions’, etc.) (Shah et al., 2021). Within the 150,377 ecogame reviews, the original dictionary identified 724 occurrences; for action/adventure game reviews, it identified only 11 occurrences.

To broaden the dictionary’s capability in capturing a more extensive array of climate change-related content, it was expanded using the model ‘gpt-4-turbo’ via the OpenAI API (<https://platform.openai.com>), tasked with augmenting the original dictionary with semantically similar terms. To make the output of the LLM more deterministic, the following parameter values were adopted: *temperature* = 0.00, *max_tokens* = 200, *seed* = 100. The *max_tokens* parameter controls the maximum length of the output, while the *temperature* and *seed* parameters influence the randomness of the model’s responses. Specifically, setting a low *temperature* value (0.00) minimizes randomness in token selection, ensuring that the model generates more predictable and focused outputs. Additionally, setting a *seed* value standardizes the

sampling process, further enhancing the consistency and reproducibility of the generated terms. Given that LLMs have proven effective in linguistic annotation tasks, with GPT-4 nearing human coder accuracy (Yu et al., 2024) and have been used to create sustainability-focused dictionaries (O’Leary & Yoon, 2023), GPT-4’s strengths were harnessed to widen the original dictionary’s narrow scope. The expanded dictionary will be henceforth referred to as the *climate-dictionary*. To prevent the inclusion of overly specific multi-word terms at the expense of linguistic frequency, the LLM was instructed explicitly to prioritize simple words common in everyday language. The specific prompt given to the LLM and its response are detailed in Appendix 1 – Table A1. As a result, the dictionary grew to include 52 unique, climate change-related terms. The 26 words created with the LLM, which expanded the original dictionary terms (Shah et al., 2021), are itemized in Table 2 along with the 26 terms of the original climate change dictionary by Shah et al. (2021). The authors acknowledge that the GPT-4-based extension may result in inconsistent term inclusions and omissions. However, the primary goal of the dictionary-based approach was not to create or validate an exhaustive dictionary but rather to broaden its scope while minimizing the need for human intervention. To assess the robustness of the approach, and to check the necessity of human oversight, additional versions of the dictionary containing 66 and 99 terms were manually created. The results of the analysis remained stable across these versions, suggesting that the findings are not dependent on the inclusion or exclusion of individual terms (see Appendix 2).

2.3.2. Climate change detection classification

In this study, we employ ClimateBERT (Webersinke et al., 2022), a domain-specific LLM from the BERT family of models (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2019), to perform climate change detection classification. During its domain-adaptive pretraining, ClimateBERT has been trained on 1.6 million climate change-related texts, including research paper abstracts, news articles, and company reports, to improve the model’s performance on the specific issue of climate change. Additionally, ClimateBERT has been fine-tuned for several downstream NLP tasks including climate change detection classification, which involves the binary classification of climate change-related paragraphs. Although the model is named ‘ClimateBERT’, its scope extends beyond a narrow interpretation of the term ‘climate change’ to include environmental topics such as energy consumption, nature, coastal area exposures, and fossil fuels (Webersinke et al., 2022). In the appendix of their paper, under the

Table 2
The original climate-dictionary (Shah et al., 2021) plus the terms added via an LLM.

Original Terms		
Carbon dioxide	Carbon emissions	Carbon footprint
Carbon tax	Ch4 emissions	Ch4 levels
Cities underwater	Climate change	Climate crisis
Climate disaster	Climate science	Climate warming
Co2	Co2 emissions	Co2 levels
El nino	Global cooling	Global warming
Greenhouse emissions	Greenhouse gases	Melting ice caps
Melting icebergs	Methane emissions	Ozone layer
Rising sea levels	Warming oceans	
Added Terms		
Air pollution	Energy efficiency	Renewable energy
Biodiversity	Fossil fuels	Solar power
Clean energy	Habitat loss	Sustainability
Conservation	Hybrid vehicles	Urban sprawl
Deforestation	Land degradation	Waste management
Drought	Natural resources	Water conservation
Eco-friendly	Organic farming	Wind energy
Ecosystem	Pollution	Zero waste
Endangered Species	Recycle	

annotation guidelines, the authors explicitly state that paragraphs are coded as “climate-related” or “climate-relevant” if they include content related to climate change or environmental topics. For specific examples of their annotations, we refer readers to the publicly available ClimateBERT climate detection classification dataset, which can be accessed on HuggingFace (https://huggingface.co/datasets/climatebert/climate_detection).

ClimateBERT has demonstrated superior performance compared to standard BERT models and various ChatGPT models on several downstream tasks (Trajanov et al., 2023; Webersinke et al., 2022). In this study, we used the HuggingFace library and the ‘climatebert/distilroberta-base-climate-detector’ model checkpoint to perform climate change detection classification (Bingler et al., 2024). The model was employed to determine, in binary terms, whether the reviews were *climate change-related* or *climate change-unrelated* (non-climate change-related). Similar to other BERT models (Devlin et al., 2019), ClimateBERT is limited to 512 tokens for each input text and reviews exceeding this limit were truncated to the first 512 tokens. The mean number of tokens in our dataset was 83.7 (*SD* = 150.2, *Mdn* = 34, *min* = 7, *max* = 3361) with only 2.27 % of reviews exceeding the 512-token limit. We posited that the first 512 tokens of a review provide a sufficiently representative summary of the entire review.

To evaluate the performance of the ClimateBERT model on our dataset we manually coded ten reviews per ecogame and calculated relevant performance metrics. To align with the classification pipeline employed in this study, only positive reviews and reviews containing more than five words were randomly selected from each ecogame, resulting in 440 manually coded reviews. All reviews were coded as 1 (climate change-related) or 0 (climate change-unrelated) by the primary author, a PhD candidate specializing in text analysis and video games, and a psychologist with a master’s degree and prior research experience in analyzing Steam reviews. In cases of disagreement, the two coders engaged in discussions until they reached a consensus, ensuring consistency and reliability in the coding process. The coding guidelines were based on the original ClimateBERT paper by Webersinke et al. (2022) and are available in Appendix 3.

2.4. Statistical analyses

For the dictionary-based approach, a Chi-Square goodness-of-fit test for a uniform distribution test was conducted using the *chi2_contingency* function from the *scipy.stats* module. This test was used to assess the frequency with which climate-dictionary terms appeared in reviews across game categories (ecogames vs. action/adventure games), thereby addressing Hypothesis 1. To analyze the results of the climate change detection classification, an independent samples *t*-test using the *ttest_ind* function from the *scipy.stats* module was employed. This test compared the levels of climate change-related content, between ecogame reviews and action/adventure game reviews, further testing Hypothesis 1. Additionally, an F-test for homogeneity of variances was calculated to evaluate Hypothesis 2 and to investigate differences in the levels of climate change-related content among ecogames compared to action/adventure games. All significance testing was performed using an alpha level of 0.05.

3. Results

In this study, the levels of climate change-related content per review were assessed via (a) the dictionary-based analysis using the LLM-extended climate-dictionary or (b) the climate change detection classification utilizing the ClimateBERT model.

3.1. Descriptive statistics

On average, reviews for action/adventure games are longer, with a mean of 60.6 words per review, than reviews for ecogames, with a mean

of 51.9 words per review. The selected ecogames and action/adventure games exhibit similar popularity – as defined by the percentage of recommendations to total reviews – with ecogames having a popularity rate of 86.3 % (min = 42.6 %, max = 98.2 %) (see Table 3) and action/adventure games having a rate of 84.8 % (min = 49.1 %, max = 97.5 %) (see Table 4). Descriptively, players also spent on average more time in-game before writing a review for ecogames ($M = 30.99h$, $Mdn = 10.74h$, $SD = 76.26h$) compared to action/adventure games ($M = 17.45h$, $Mdn = 15.00h$, $SD = 13.64h$) (see Tables 3 and 4). Besides, the total average playtime was higher for ecogames ($M = 58.98h$, $Mdn = 14.31h$, $SD = 157.34h$) than for action/adventure games ($M = 33.40h$, $Mdn = 24.93h$, $SD = 28.40h$). However, in both cases, the respective averages are not the best measures to capture typical tendencies due to the substantial influence of outliers, see Tables 3 and 4. In that case, the medians are better suited to represent typical tendencies. In both cases, the median is descriptively higher for action/adventure games.

As discussed in the discussion section (subsection 4.4.), both the *playtime at review* and the game's *popularity* are important general game

Table 3

Playtimes (in h) at review and popularity per ecogame.

Game	Playtime at review			Popularity
	Mean	SD	Median	
Plasticity	0.64	1.31	0.43	80.7 %
Trash Time	1.34	3.25	0.51	95.8 %
Survival Tycoon	1.67	2.95	0.83	55.6 %
Your Future Self	2.18	1.63	1.88	80.5 %
Koral	2.59	2.03	2.38	80.1 %
Flower	2.94	5.75	1.75	78.8 %
The Unclearness	2.95	2.21	1.98	61.1 %
Gibbon: Beyond the Trees	3.23	6.33	1.49	91.2 %
The Eternal Castle [REMASTERED]	3.56	6.69	2.68	82.4 %
Beecarbonize	3.67	5.52	2.0	91.6 %
Alba: A Wildlife Adventure	4.65	5.63	3.88	98.2 %
Preserve	4.87	7.75	2.48	96.4 %
1000 days to escape	4.95	9.78	2.47	82.8 %
The Aquatic Adventure of the Last Human	5.61	4.31	4.97	74.3 %
Lumino city	5.71	4.4	5.32	85.3 %
Bionic Heart	5.81	15.73	3.58	72.9 %
Endling - Extinction is forever	5.97	9.28	4.44	95.7 %
UNDER the SAND - a road trip game	6.28	22.45	3.23	69.6 %
Greed Corp	9.64	15.85	5.97	85.8 %
A new beginning - final cut	10.54	8.29	10.09	64.0 %
Terra Nil	10.93	14.64	7.73	86.0 %
Block'hood	11.09	45.7	4.93	70.3 %
Penguins Arena: Sedna's World	11.13	114.08	1.5	91.1 %
Forged Battalion	11.45	38.3	5.43	52.9 %
The Descendant	12.61	87.71	4.03	87.6 %
Democracy 3	13.49	27.04	6.61	74.8 %
Redout: Enhanced Edition	15.22	46.73	5.5	85.8 %
Dark Future: Blood Red States	15.97	151.86	4.08	60.7 %
Bear and Breakfast	17.87	31.59	11.03	90.7 %
Ziro	18.07	39.13	1.34	64.0 %
President for a Day - Corruption	23.24	142.27	0.78	42.6 %
Fate of the World: Tipping Point	28.08	50.28	11.44	77.9 %
Project Aura	34.05	81.57	9.67	58.6 %
Mad Max	39.72	66.39	29.88	91.6 %
Anno 2205	45.99	142.45	21.1	68.8 %
Niche - a genetics survival game	62.04	154.4	15.48	88.4 %
Anno 2070	72.1	153.36	24.0	55.6 %
Power & Revolution 2021 Edition	77.8	129.11	33.16	55.7 %
Eco	166.16	422.95	41.1	79.3 %
Arma 3	462.34	967.2	101.53	91.2 %
Cities: Skylines - Green Cities	*			79.4 %
Civilization VI: Gathering Storm	*			66.2 %
The Sims 4: Eco Lifestyle	*			63.3 %
Tropico 5 Waterborne DLC	*			72.9 %

Note: Some games (*) do not contain information about playtime because they are Downloadable Content (DLCs), which expands upon a pre-existing game and can be purchased separately. These tend to be games with high average playtimes at review (e.g., *Sims*, *Civilization*, etc.). Popularity is defined as the percentage of recommendations to total reviews.

Table 4

Playtimes (in h) at review and popularity per action/adventure game.

Game	Playtime at review			Popularity
	Mean	SD	Median	
PILGRIM	4.24	5.21	3.42	84.2 %
VLADIK BRUTAL	5.62	6.19	4.47	89.5 %
Turok 3: Shadow of Oblivion Remastered	5.81	5.96	4.32	86.4 %
Turnip Boy robs a bank	6.01	4.0	5.63	95.8 %
Planet of Lana	6.09	4.47	5.45	93.9 %
Unravel Two	6.55	14.21	5.08	77.0 %
Compound	7.38	12.48	3.75	97.5 %
Senua's Saga: Hellblade II	7.43	5.16	7.25	85.4 %
Arizona Sunshine 2	77.76	33.46	5.97	80.1 %
Mortal Sin	8.89	12.68	4.28	96.9 %
Scars Above	8.19	8.48	7.45	76.5 %
Battle Shapers	9.18	16.75	4.71	85.1 %
Observer: System Redux	9.68	7.75	8.65	85.4 %
The Entropy Centre	9.87	10.56	9.15	89.3 %
Momodora: Moonlight Farewell	9.96	6.78	9.12	94.1 %
Star Trek: Resurgence	10.11	7.09	10.33	89.8 %
Welcome to Paradise	11.13	14.46	6.17	70.7 %
Expeditions: A Mudrunner Game	12.46	20.03	5.93	49.4 %
Wizard with a Gun	13.06	13.6	9.55	83.8 %
METAL GEAR SOLID 3: Snake Eater - Master Collection Version	13.28	29.71	9.17	82.4 %
Goat Simulator 3	13.85	32.46	7.21	97.5 %
Chorus	15.00	12.59	14.3	83.7 %
Rayman Origins	15.65	52.23	8.68	93.8 %
Another Crab's Treasure	16.05	12.65	15.68	95.9 %
NARUTO X BORUTO Ultimate Ninja STORM CONNECTIONS	16.29	46.69	5.99	49.1 %
Le Mans Ultimate	17.08	40.84	5.27	70.3 %
Prince of Persia The Lost Crown	17.83	16.03	15.35	84.6 %
Freedom Planet	18.06	91.55	8.33	94.9 %
Crash Bandicoot 4: It's About Time	18.80	29.96	8.73	82.9 %
Lorelei and the Laser Eyes	19.54	12.4	18.53	95.9 %
Forever Skies	19.56	31.71	11.38	88.3 %
TerraTech Worlds	20.14	44.13	7.9	71.6 %
WRC Generations	20.46	65.39	6.53	63.1 %
Ember Knights	21.73	34.77	11.48	95.1 %
Grand Theft Auto: San Andreas - The Definitive Edition	22.09	33.23	11.11	71.5 %
SCARLET NEXUS	24.39	29.65	16.32	85.3 %
Moonstone Island	26.30	32.69	14.3	87.1 %
Far Cry 6	28.99	77.73	16.08	67.6 %
Dwarven Realms	29.45	99.43	9.23	86.7 %
ASKA	32.01	53.15	15.3	79.2 %
Eiyuden Chronicle: Hundred Heroes	36.85	38.31	23.32	84.2 %
Assassins Creed Valhalla	45.93	88.41	18.37	67.2 %
Assassins Creed Odyssey	82.42	164.08	51.52	89.7 %
Kingdom Come: Deliverance – Band of Bastards	*			68.8 %
The Talos Principle 2 – Road to Elysium	*			95.5 %

Note: Some games (*) do not contain information about playtime because they are DLCs. These tend to be games with high average playtimes at review (e.g., *Sims*, *Civilization*, etc.). Popularity is defined as the percentage of recommendations to total reviews.

metrics for gauging suitability as educational tools. We examine playtime at the time of review as a proxy for engagement with the game, rather than total playtime, which can be open-ended for some games. The playtime at the review presumably reflects how long players engaged with the game before feeling equipped to provide an informed opinion on its content and mechanics. This engagement can vary significantly between different genres, indie and non-indie games, and free-to-play and non-free-to-play games (Lin et al., 2019).

3.2. Dictionary-based approach

A Chi-Square goodness-of-fit test for a uniform distribution was performed to compare the frequency of terms from the climate-dictionary observed in ecogames versus action/adventure games. The

terms appeared 2015 times within ecogame reviews, whereas only 210 instances were found in action/adventure game reviews (see Fig. 2). This significant disparity, $\chi^2(1, N = 2225) = 1678.77, p < .001$, suggests that climate change-related terminology is more commonly associated with ecogames reviews.

The top five most frequently occurring climate-dictionary terms in ecogame reviews were: ‘pollution’ ($n = 326$), ‘ecosystem’ ($n = 322$), ‘climate change’ ($n = 288$), ‘global warming’ ($n = 155$) and ‘conservation’ ($n = 126$) (see Fig. 2). Comprehensive frequency counts for all games with more than zero instances of these terms are presented in Table 5. The review with the highest occurrence of these terms ($n = 11$) is provided in Appendix 4 – Table A5.

3.3. Climate change detection classification

This study employs the domain-specific ClimateBERT model to classify reviews into one of two categories: *climate change-related* or *climate change-unrelated*. The underlying RoBERTa model was pretrained on climate change-related texts, including news articles, research paper abstracts, and company reports, and subsequently fine-tuned on several downstream tasks, such as the detection of climate change-related paragraphs (Webersinke et al., 2022). Since the model was not trained on social media text, we first evaluated its performance on a manually labeled dataset of 440 randomly selected ecogame reviews.

3.3.1. Climate change detection classification performance

In the first round of coding, the two coders achieved a Cohen’s Kappa of 0.86. After resolving inconsistencies through further discussion, the manually coded dataset was compared to the model’s predictions. The ClimateBERT model achieved an accuracy score of 0.91, a precision score of 0.74, a recall score of 0.77, and an F1 score of 0.75 (see Table 6).

Cohen’s Kappa may be a more suitable performance metric for binary classifiers in cases where class imbalances exist, as in the present study (Christen et al., 2024; Powers, 2015) (see Table 6). The Cohen’s Kappa between the humanly-coded ‘ground truth’ and the predictions generated by ClimateBERT was 0.70, indicating a ‘substantial agreement’ (Landis & Koch, 1977).

3.3.2. Climate change detection classification results

The category ‘climate change-unrelated’ was the most prevalent category for both ecogames ($M = 84.72\%$; $SD = 16.48\%$) and action/

adventure games ($M = 96.92\%$; $SD = 2.91\%$), suggesting that the vast majority of discourse on Steam is unrelated to climate change.

An independent samples t -test compared the relative frequency of the ‘climate change-related’ category, revealing that this category is discussed significantly more in ecogame reviews ($M = 15.28\%$; $SD = 16.48\%$) than in action/adventure game reviews ($M = 3.07\%$; $SD = 2.91\%$) ($t(87) = 4.89, p < .001$, Cohen’s $d = 1.03$) (see Fig. 3). The relative frequency indicates the proportion of reviews categorized as ‘climate change-related’ relative to the total number of available reviews for that game. Despite its greater occurrence in ecogame, climate change-related content still reflects only a secondary focus, comprising just 15.28 % of total discussion. In addition to the binary classification, the ClimateBERT model also outputs a probability score for the assigned label (ranging from 0.5 to 1). Reviews with the highest ‘climate change-related’ scores – all originating from ecogames – are presented in Table 7.

The dispersion of the climate change-related content in reviews of ecogames is significantly wider (0.0 %–67.9 %) than in action/adventure games (0.3 %–14.3 %), as in agreement with an F-test for homogeneity of variances ($F(43,44) = 23.92, p < .001$) (see Fig. 3). Detailed scores for the ‘climate change-related’ category of each game are available in Table 8.

3.4. Potential to identify ‘new’ ecogames

This study randomly selected 45 action/adventure games as a control group to compare levels of climate change-related content with ecogames. Consequently, it was possible that the action/adventure games included in this study also incorporate environmental topics and feature climate change-related content to varying degrees.

Some action/adventure games, such as *Another Crab’s Treasure* (Another Crab’s Treasure, 2024), *ASKA* (ASKA, 2024), and *Forever Skies* (Forever Skies, 2025) ranked among the top action/adventure games in both the dictionary-based analyses and the climate change detection classification, suggesting they might qualify as ecogames. *Another Crab’s Treasure* was the action/adventure game with the highest total count of climate change-related terms and the only action/adventure game with counts in the triple digits ($n = 101$). Moreover, it is the most-liked game out of the three with a popularity of 96 % (see Table 4). Below are some sample rephrased reviews of the game containing the terms in the pre-defined climate-dictionary:

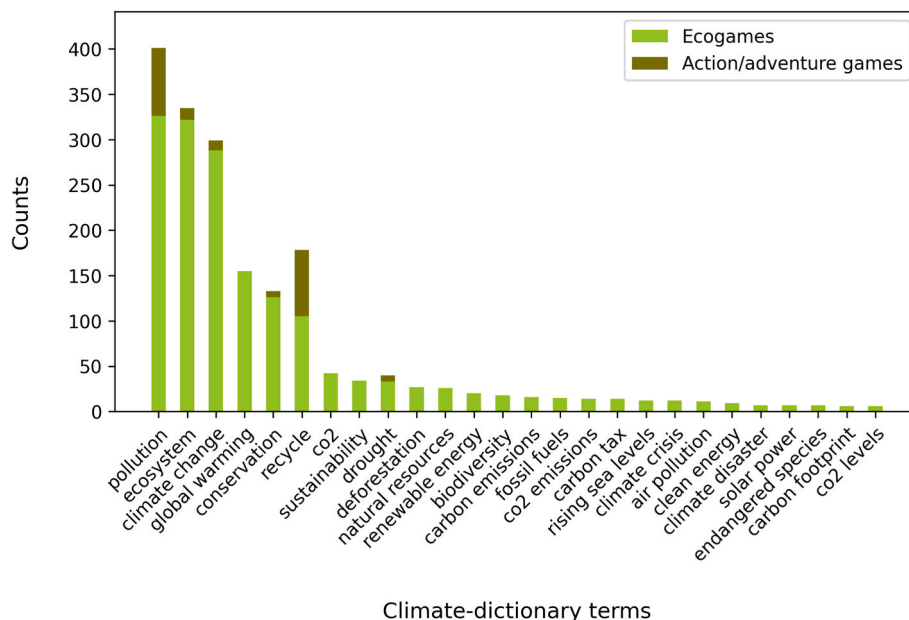


Fig. 2. Barplot showing the absolute frequencies of climate-dictionary terms ($n > 5$) in ecogame reviews compared to action/adventure game reviews.

Table 5

Counts of climate-dictionary terms for ecogame reviews and action/adventure game reviews.

Ecogames			Action/Adventure Games		
Title	Rel. freq.	Counts	Title	Rel. freq.	Counts
The Sims 4: Eco Lifestyle	0.355	11	Another Crab's Treasure	0.014	101
Fate of the World: Tipping Point	0.341	101	Battle Shapers	0.012	4
Civilization VI: Gathering Storm	0.284	286	TerraTech Worlds	0.006	4
Koral	0.178	23	NARUTO X BORUTO Ultimate Ninja STORM CONNECTIONS	0.006	4
Beecarbonize	0.177	169	Forever Skies	0.006	15
A new beginning - final cut	0.17	89	The Talos Principle 2 - Road to Elysium	0.006	2
Cities: Skylines - Green Cities	0.145	28	Scars Above	0.005	4
Project Aura	0.106	27	WRC Generations	0.004	3
Terra Nil	0.093	290	Kingdom Come: Deliverance - Band of Bastards	0.004	1
Plasticity	0.091	27	Turok 3: Shadow of Oblivion Remastered	0.003	1
Trash Time	0.087	2	Wizard with a Gun	0.002	3
Block'hood	0.087	39	SCARLET NEXUS	0.002	9
Gibbon: Beyond the Trees	0.086	25	Far Cry 6	0.001	9
Eco	0.068	406	Expeditions: A Mudrunner Game	0.001	1
Tropico 5 Waterborne DLC	0.057	2	Ember Knights	0.001	4
Alba: A Wildlife Adventure	0.04	117	Chorus	0.001	2
Preserve	0.038	5	ASKA	0.001	2
Bionic Heart	0.032	2	Planet of Lana	0.001	2
Endling - Extinction is forever	0.032	34	Moonstone Island	0.001	2
Anno 2070	0.025	100	PILGRIM	0.001	1
Democracy 3	0.025	99	Dwarven Realms	0.001	1
Power & Revolution 2021 Edition	0.019	5	Eiyuden Chronicle: Hundred Heroes	0.001	1
The Aquatic Adventure of the Last Human	0.017	4	Freedom Planet	0.001	2
Your Future Self	0.015	1	Assassins Creed Odyssey	0.001	28
Penguins Arena: Sedna's World	0.015	12	The Entropy Centre	0.0	1
Greed Corp	0.012	4	Mortal Sin	0.001	1
UNDER the SAND - a road trip game	0.009	4	Rayman Origins	0.001	1
Bear and Breakfast	0.009	27	Assassins Creed Valhalla	0.001	1
Dark Future: Blood Red States	0.008	2			
Flower	0.004	6			
Anno 2205	0.004	8			
Niche - a genetics survival game	0.003	12			
The Eternal Castle [REMASTERED]	0.003	1			
Lumino city	0.003	2			
Forged Battalion	0.003	1			
Mad Max	0.001	28			
The Descendant	0.0	1			
Redout: Enhanced Edition	0.0	1			
Arma 3	0.0	14			

Note: Games with zero counts are not included in this table. The relative frequency (rel. freq.) represents the counts of climate-dictionary terms in a game's reviews relative to the total number of reviews for that game.

Table 6

Confusion matrix of the climate change detection classification performance comparing ClimateBERT to the human coders.

True Class	Predicted Class		
	0	1	2
0	341	21	0
1	18	60	1
2	0	1	0

Note: The category 'climate change-unrelated' was coded as 0 and the category 'climate change-related' was coded as 1.

'A stunningly beautiful game that brought me to tears by the end. If the touching narrative of this game doesn't awaken a sense of urgency in you about the importance of combating climate change and preserving our oceans, I can't imagine what will.' – review of *Another Crab's Treasure* ([Another Crab's Treasure, 2024](#))

'I know there's a lot of backlash when games tackle issues like the pitfalls of capitalism or the realities of global warming, but this game absolutely delivers it right. It's astonishing how well-done these themes are; you'd think it was created by a big studio, not this small team. I've been anticipating this game's release and it exceeded my expectations. Easily one of my all-time favorites, and I can't await this team's next project.' – review of *Another Crab's Treasure* ([Another Crab's Treasure, 2024](#))

'Man, this game is gorgeous. it's my first soulslike and this was a perfect intro to the genre; when you finally figure out the combat for a boss or enemy, it feels amazing. The progression feels super rewarding and the challenges are tough but totally doable. the environment is stunning and the combat is so satisfying, plus there's always a reason to explore and go back to old areas. i really love how it focuses on pollution and trash; you can't go too far without getting hit with the reality of our oceans, and it kinda sticks with you. maybe it even makes you wanna do something about it in real life?! Just an awesome game and totally worth your time!!' – review of *Another Crab's Treasure* ([Another Crab's Treasure, 2024](#))

If *Another Crab's Treasure* were considered among ecogames, it would rank 26th out of 46 based on the relative frequency of climate-dictionary terms in reviews according to the dictionary analysis. Furthermore, it would rank 28th out of 46 based on the climate change detection classification in terms of the relative frequency of the 'climate change-related' category. These findings highlight that previously overlooked action/adventure games may, in fact, incorporate significant environmental topics.

4. Discussion

This study evaluated the level of climate change-related content in reviews of ecogames on the Steam gaming platform and compared it to reviews of action/adventure games using dictionary-based analysis and climate change detection classification. Furthermore, the dispersion of climate change-related content across different ecogames was examined and potential new ecogame candidates were explored.

4.1. Main findings

Our dictionary-based analyses suggested that predefined climate-related terms were significantly more present in ecogames reviews. This finding was corroborated by the climate change detection classification results, which also indicated a higher prevalence of climate change-related content in ecogame reviews compared to action/

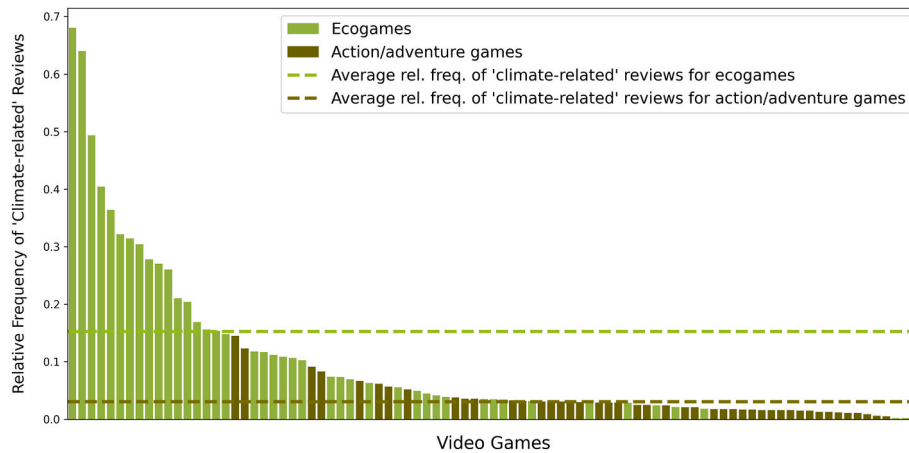


Fig. 3. Barplot of climate change detection classification results for ecogame reviews compared to action/adventure game reviews. Each bar represents a video game, with the relative frequency indicating the proportion of reviews categorized as 'climate change-related' relative to the total number of available reviews for that game.

adventure game reviews. Thus, Hypothesis 1, which posited a higher level of climate change-related content in ecogames, was supported by the consistent findings from both analytical approaches. This underscores the capability of the NLP pipeline used in this study to distinguish between ecogames and non-ecogames and appraise the extent of climate change-related content by examining player-generated reviews of the games. Moreover, the results lend credibility to the ecogame category as a valid thematic classification, distinct from an established genre, and align with the list of games proposed by Fernández Galeote and Hamari (2021). Even though some ecogames may be scattered among other genres, as demonstrated in this study with *Another Crab's Treasure* (Another Crab's Treasure, 2024) being categorized as an action/adventure game, ecogames remain a statistically distinguishable category. This reinforces the validity and utility of the ecogame classification for both research and educational purposes.

Additionally, our analyses revealed significant variations in the extent of climate change-related content within reviews of ecogames, supporting Hypothesis 2. Accordingly, not all ecogames might be equally effective in facilitating environmental discussions in team-based debriefing sessions after gameplay. As such, certain games might be more/less suitable for educational and/or research purposes. This variability can partly be attributed to the broad definition of 'ecogame' which subsumes a diverse range of video games. For instance, based on existing literature and a list of video games representing climate change (Fernández Galeote & Hamari, 2021), the present study classified e.g., *Arma 3* (Arma 3., 2013) and *Redout: Enhanced Edition* (Redout: Enhanced Edition, 2016), and *Mad Max* (Mad Max, 2015) as ecogames, despite their less pronounced focus on environmental topics compared to titles like *Beecarbonize* (Beecarbonize, 2023) and *Civilization VI: Gathering Storm* (Civilization VI: Gathering Storm, 2019). This underscores the importance of selecting specific, content-related ecogames based on data-driven criteria for educational and/or research usage rather than relying on random selection. Additionally, future studies exploring psychological engagement with climate change could benefit from leveraging the affective (review sentiment), cognitive (attitude), and behavioral (playtime) dimensions of the data corpus generated in this study. These aspects offer valuable insights into how players interact with and respond to climate change-related content in games.

Importantly, the current NLP-based approach provides a data-driven analysis of climate change-related content in games. This could substantially simplify the selection process for educators who aim to incorporate games into their curriculum to promote environmental education, even for those with limited knowledge of gaming or ecogames specifically. Consequently, our approach could serve as a valuable tool for educators to narrow down potential games for a more detailed

examination, assessing their appropriateness for classroom use in stimulating discussions on environmental issues in team-based debriefings. Moreover, this approach can benefit the research community by introducing structure into the often-challenging game selection process for studies, thereby enhancing the overall quality of future research (Tyack et al., 2018).

Unsurprisingly, the climate change detection classification revealed that most discussions in the reviews were predominantly unrelated to climate change, falling into the 'climate change-unrelated' category. Given that gaming-related subjects, such as user experience, narrative, and game design (Zagal et al., 2009), dominate conversations about games, this outcome is understandable. Climate change-related content was discussed in only 15.28 % of the ecogame reviews, reiterating its secondary nature within the broader context of video game discourse. Despite the limited number of reviews featuring climate change-related content – even among ecogames – our comprehensive analysis indicates that NLP techniques are valuable tools for pinpointing ecogames that feature high-levels of discussions on environmental issues in their reviews. While working with social media data remains a 'messy' undertaking (Baldwin et al., 2013; Bendle, Wang & Shane, 2016; Kumar & Khosla, 2018), NLP and LLMs may offer effective solutions for managing this vast amount of unstructured text data. The NLP pipeline presented here provides a fast, scalable, and data-driven method for preselecting games, beneficial for educators and researchers alike. Accordingly, procuring specific and topical video games for educational or laboratory use becomes easier, especially for non-experts, even amidst thousands of potential titles. Moreover, the NLP methodology employed in this study has the potential to inform the design of future "serious" ecogames by identifying key themes and player responses that resonate with audiences. At the same time, the methodology offers a means of evaluating the effectiveness of such games by analyzing the content of associated player reviews. This dual utility underscores the broader applicability of the approach in both educational and game development contexts.

4.2. ClimateBERT and steam reviews

The ClimateBERT classifier demonstrated adequate performance on our dataset, achieving an F1 score of 0.75 and a Cohen's Kappa of 0.7. However, this F1 score is notably lower than the F1 score of 0.989 reported in the original study (Webersinke et al., 2022). The performance drop was expected and can largely be attributed to the unstructured and 'messy' nature of social media text (Baldwin et al., 2013; Eisenstein, 2013). Reviews on Steam often include ASCII drawings/art, in-side jokes, specific gaming slang, satire, and hyperbole (Phillips et al., 2021) which complicates text analysis. In contrast, the F1 reported in

Table 7

Top three reviews with the highest ‘climate change-related’ scores.

Game	Score	Review
Beecarbonize	0.998	Beecarbonize is a free card game all about tackling climate change and cutting down carbon emissions. The goal is to keep a balance between things like industry, social reforms, eco-friendly policies, and scientific progress to stop an environmental disaster. Gameplay is centered on using cards that represent real-world climate ideas and making smart choices to manage your resources. The cool part is the game’s world responds to what you do, so your actions trigger all kinds of results, both good and bad.
Beecarbonize	0.998	“Beecarbonize” is a card strategy game that’s been turning heads on Steam because of its fresh take on teaching players about climate change. Made by Charles Games, it’s got a ‘Very Positive’ rating and people are loving both the gameplay and the way it gets you thinking about the environment. Players are calling “Beecarbonize” a great way to learn about practical climate solutions while having fun. It’s like an undercover educational tool that shines a light on conservation and humanity’s effect on the planet. Here’s what makes “Beecarbonize” worth checking out: • Strategic Gameplay: You’ll be researching tech, passing policies, saving ecosystems, and upgrading industries to cut carbon emissions. • Grounded in Science: The game’s cards are based on real climate science—they represent actual inventions, policies, industries, or social changes. • Different Paths to Win: Whether you’re reforming industries, protecting nature, or working on community initiatives, there’s more than one way to succeed. • Changing World Events: Random events shake things up, so you’ve gotta adjust your plans to deal with whatever happens. • Hardcore Mode: Feeling confident? Hardcore mode cranks up the difficulty, showing just how tough the climate crisis can get.
Eco	0.998	If you’re into strategic games and want to dive deeper into environmental issues in an interactive way, “Beecarbonize” could totally be the game for you. It’s a fun, thoughtful challenge that lets you understand the forces shaping our world. Plus, it’s completely free! This is a great game for a group of friends who want to feel like they’re heading into the wilderness to build their own little society. Starting out with only a rough idea of what to do keeps you motivated to take on the tasks you choose for yourself. Plus, it really pushes you to think about using resources more sustainably and finding the right balance. Honestly, it feels like a solid reminder of how climate change impacts the world we live in and why we need to approach things with care for the planet.

Note: Reviews were rephrased to avoid copyright infringement. The scores represent the probability, as indicated by ClimateBERT, that the review falls into the ‘climate change-related’ category.

the original study by [Webersinke et al. \(2022\)](#) was achieved by classifying company reports. While ClimateBERT has been specifically adapted to the domain of climate change discourse, it has not been fine-tuned for the particular context of ‘video game language’ in social media contexts, which could partially explain the classifier’s decreased performance.

Furthermore, ClimateBERT may have been challenged by games that reference environmental topics in their titles, such as *Endling – Extinction is Forever* ([Endling - Extinction is Forever, 2022](#)), *Eco* ([Eco, 2018](#)), *The Sims 4 Eco Lifestyle* ([The Sims 4 Eco Lifestyle, 2020](#)), or *Cities: Skylines – Green Cities* ([Cities: Skylines - Green Cities, 2017](#)). The scores for these games may be inflated due to misclassifications into the ‘climate change-related’ category based solely on the titles. Although we did not systematically evaluate the influence of game titles on classification outcomes, we conducted a quick quality check by randomly selecting 10 reviews for the game *Eco* ([Eco, 2018](#)) containing the title of the game.

Table 8

The climate change detection classification results for ecogame and action/adventure game reviews.

Ecogames			Action/Adventure Games		
Title	n	Rel. freq.	Title	n	Rel. freq.
The Sims 4: Eco Lifestyle	28	0.679	Expeditions: A Mudrunner Game	636	0.143
Cities: Skylines - Green Cities	163	0.638	TerraTech Worlds	528	0.121
Koral	118	0.492	Forever Skies	2123	0.089
Fate of the World: Tipping Point	266	0.402	WRC Generations	558	0.081
Beecarbonize	743	0.362	NARUTO X BORUTO Ultimate Ninja STORM CONNECTIONS	416	0.065
Civilization VI: Gathering Storm	832	0.32	ASKA	1568	0.06
Trash Time	16	0.312	Another Crab’s Treasure	5095	0.055
Plasticity	255	0.302	VLADiK BRUTAL	760	0.05
Tropico 5 Waterborne DLC	29	0.276	Goat Simulator 3	1200	0.036
Terra Nil	2676	0.268	Scars Above	787	0.034
A new beginning - final cut	488	0.258	Welcome to Paradise	175	0.034
Block’hood	413	0.208	Observer: System Redux	829	0.031
Alba: A Wildlife Adventure	2568	0.202	Planet of Lana	1555	0.03
Eco	4536	0.167	Wizard with a Gun	1139	0.029
Ziro	13	0.154	Grand Theft Auto: San Andreas - The Definitive Edition	863	0.029
Gibbon: Beyond the Trees	231	0.152	The Entropy Centre	1560	0.029
Project Aura	240	0.146	Mortal Sin	1690	0.028
Endling - Extinction is forever	899	0.116	Crash Bandicoot 4: It’s About Time	888	0.028
Preserve	122	0.115	Far Cry 6	3881	0.027
The Aquatic Adventure of the Last Human	209	0.11	METAL GEAR SOLID 3: Snake Eater - Master Collection Version	1380	0.027
UNDER the SAND - a road trip game	328	0.107	Le Mans Ultimate	1046	0.027
Flower	1120	0.104	Ember Knights	2400	0.023
Anno 2070	3423	0.1	Assassins Creed Valhalla	3951	0.023
1000 days to escape	111	0.072	PILGRIM	1019	0.022
Bionic Heart	56	0.071	Unravel Two	1385	0.019
Mad Max	25101	0.068	Senua’s Saga: Hellblade II	1870	0.019
Anno 2205	1569	0.061	Moonstone Island	1651	0.016
Penguins Arena: Sedna’s World	598	0.054	Prince of Persia The Lost Crown	509	0.016
Forged Battalion	338	0.047	The Talos Principle 2 - Road to Elysium	319	0.016
Bear and Breakfast	2459	0.043	Compound	1522	0.015
Redout: Enhanced Edition	2328	0.04	Arizona Sunshine 2	596	0.015
Dark Future: Blood Red States	217	0.037	Assassins Creed Odyssey	31557	0.014
The Descendant	631	0.033	Momodora: Moonlight Farewell	733	0.014
Greed Corp	306	0.033	Rayman Origins	1613	0.014
Democracy 3	3567	0.032	Dwarven Realms	1259	0.014
The Eternal Castle [REMASTERED]	304	0.03	SCARLET NEXUS	4033	0.013
Niche - a genetics survival game	3104	0.027	Star Trek: Resurgence	900	0.013

(continued on next page)

Table 8 (continued)

Ecogames			Action/Adventure Games		
Title	n	Rel. freq.	Title	n	Rel. freq.
Lumino city	642	0.026	Turok 3: Shadow of Oblivion Remastered	263	0.011
Power & Revolution 2021 Edition	181	0.022	Freedom Planet	3063	0.011
Arma 3	33260	0.02	Battle Shapers	300	0.01
Your Future Self	62	0.016	Turnip Boy robs a bank	951	0.009
Survival Tycoon	12	0.0	Lorelei and the Laser Eyes	637	0.009
The Uncleaness	11	0.0	Chorus	1607	0.007
President for a Day - Corruption	15	0.0	Kingdom Come: Deliverance - Band of Bastards	233	0.004
Bob & Kuura: Lost in Snowglobe	0	NaN	Eiyuden Chronicle: Hundred Heroes	1363	0.003

Note: The relative frequency indicates the proportion of reviews categorized as ‘climate change-related’ relative to the total number of available reviews for that game.

This game was chosen as the title provides a concise and clear reference to environmental topics. We classified short excerpts (1–2 sentences) from the reviews that did not reference climate change-related topics but included the title (e.g., ‘Eco is one of the most captivating yet infuriating games I’ve ever experienced.’, ‘Eco is a fantastic game! It’s like a grown-up version of Minecraft, focusing on tech advancement and teamwork.’, ...). In all ten cases, ClimateBERT accurately assessed the reviews as ‘climate change-unrelated’. Additionally, the results of the climate change detection classification and the dictionary analyses converge for most titles, lending further credibility to the classification results. *Eco* (Eco, 2018) ranks 14th in both NLP approaches, *Endling – Extinction is Forever* (Endling - Extinction Is Forever, 2022), ranks 18th for the climate change detection classification and 19th for the dictionary analysis, *Skylines – Green Cities* (Cities: Skylines - Green Cities, 2017) ranks 2nd for the climate change detection classification and 7th for the dictionary analysis, and *The Sims 4 Eco Lifestyle* (The Sims 4 Eco Lifestyle, 2020) ranks 1st in both approaches. While some bias from game titles may exist nonetheless, the authors are confident that it does not significantly impact the outcomes of this study. Overall, the ClimateBERT model performed reasonably well, particularly given the inherent challenges associated with analyzing unstructured data.

4.3. Identifying ‘new’ ecogames

Another significant finding of this study is that the NLP methodology is adept at uncovering previously unrecognized games, such as *Another Crab’s Treasure* (Another Crab’s Treasure, 2024), *ASKA* (ASKA, 2024), *Forever Skies* (Forever Skies, 2025), as potential ecogames. Those titles, which rank among the highest in the action/adventure category, would have been classified as mediocre ecogames, outperforming the lower-ranking ecogames such as *Arma 3* (Arma 3., 2013) and *Redout* (Redout: Enhanced Edition, 2016). Positioned somewhere in the midfield of ecogames, these titles may not be prime candidates for classroom use or research studies.

Notably, this study randomly selected only 45 action/adventure, suggesting that applying the NLP methodology to all 133,997 games on Steam (SteamDB, n.d.) could reveal even-more high-ranking and promising ecogames. The method’s ability to keep pace with the fast-moving gaming market is a distinct advantage, allowing for the inclusion of newer games that traditional research methods would likely overlook. For instance, all three mentioned potential ecogames were released in 2024 or 2025 and would likely not have been discovered

using slower, conventional research processes.

4.4. From NLP results to a practical example of selecting games

Building on our findings, we now illustrate the method by which a teacher or researcher might identify a game suitable for fostering discussions on environmental issues during team-based debriefings after gameplay. While the implementation of this methodology is technically feasible for motivated educators – especially as we have made the relevant scripts publicly available on OSF – we anticipate that researchers will take the lead in curating shortlists of games for classroom or research purposes. Beyond the content of reviews reflecting climate change-related content as identified with our NLP approach, other general game metrics can, of course, also aid in gauging the appropriateness of a game for pedagogical or research purposes.

A game’s *popularity*, indicated by percentage of recommendations to total reviews, is an insightful variable providing a quick overview of how a game generally appeals to an audience. Games with high levels of climate change-related content that are widely disliked may not be suitable as educational tools.

Playtime is also a crucial factor, as educational and laboratory constraints limit the feasibility of certain games. Games that require extensive playtime to grasp complex mechanics, such as *Fate of the World: Tipping Point* (Fate of the World: Tipping Point, 2011) or *Civilization VI: Gathering Storm* (Civilization VI: Gathering Storm, 2019), are usually unsuitable. Besides, gaming sessions in classrooms or research laboratories are often followed by debriefing sessions to facilitate deeper processing of the information, which requires extra time (Crookall, 2014). Conversely, games offering briefer experiences, like *Plasticity* (Plasticity, 2019) and *Koral* (Koral, 2019) may be better suited for classroom or laboratory contexts. If duration is not a limitation and longer-term gaming sessions are preferred – perhaps spread over the course of a school year or in informal learning settings – longer games can be considered. For example, *Eco* (Eco, 2018), with an average playtime of 166.16 h at review, is one such title. Table 3 lists average playtimes and popularity metrics for each ecogame.

Age rating is another important consideration. Games like *Mad Max* (Mad Max, 2015) with an age rating of 18, are not suitable for minors but could be appropriate for university students. Steam also details each game’s technical requirements.

By prioritizing shorter gaming experiences and considering popularity, along with the NLP metrics, this study can suggest several ecogames for further exploration and educational incorporation. Steam’s most popular user-defined tags per game offer quick insights into the content of the recommended titles. *Beecarbonize* (Beecarbonize, 2023) (tags: Strategy, Resource Management, Card Game, Simulation, Nature) ranks highly in both NLP domains but has a slightly longer gameplay duration than *Gibbon: Beyond the Trees* (Gibbon: Beyond the Trees, 2022) and somewhat less popularity than *Alba: A Wildlife Adventure* (Alba: A Wildlife Adventure, 2020). As the top five tags suggest, *Beecarbonize* is a strategic, card-based simulation of climate change, while *Alba: A Wildlife Adventure* (tags: Casual, Wholesome, Cozy, Family Friendly, 3D) offers a wholesome, short gameplay experience focused on exploration and wildlife conservation. *Gibbon: Beyond the Trees* (tags: Story Rich, Adventure, Well-Written, Atmospheric, Soundtrack) is a short, emotional, narrative-driven game about habitat destruction affecting gibbons. Depending on the situational demand and the objectives of the research or the teaching unit, different games may be chosen. The NLP-driven approach is not intended to replace expert or human evaluation but rather to support it by narrowing down the pool to a smaller number of candidate games. The final decision regarding implementation in classrooms or research will always rest with a human evaluator. For example, *Gibbon: Beyond the Trees*, with an age rating of 12, may be too emotionally stirring for younger audiences, whereas *Alba: A Wildlife Adventure*, with an age rating of 3, may be more appropriate in such contexts. *Beecarbonize*, as a card-based simulation of climate change and

arguably a simpler, more modern version of *Fate of the World: Tipping Point* (*Fate of the World: Tipping Point*, 2011) may be particularly suitable for targeting the development of systems thinking (Waddington & Fennewald, 2018). Table 9 spotlights the top three recommended ecogames, each with individual strengths and considerations for various contexts. While empirical studies examining the effects of these games on environmental attitudes and awareness are still needed, some research has hinted towards the potential of the top three games highlighted in this study (Larreina-Morales & Gunella, 2023; Lu et al., 2023; Steinhauser et al., 2024).

Other noteworthy games should also be mentioned: *Trash Time* (*Trash Time*, 2019) was not included as one of the top three recommendations despite its short playtime, high popularity, and high climate change detection classification score, as it only has 24 reviews, which impedes meaningful analysis. To ensure the robustness of our results, we chose to explicitly include games in the recommendations only if they have 100 or more reviews. *Koral* (Koral, 2019) and *Plasticity* (Plasticity, 2019) demonstrated very short playtimes and high scores on the NLP analyses, but comparatively low popularity ratings of 81 % and 80 %. In contrast, *Endling: Extinction is forever* (Endling - Extinction Is Forever, 2022) and *Preserve* (Preserve, 2024) exhibited high popularity scores and relatively short playtimes, though they ranked slightly in the NLP analyses. Nevertheless, these games may still be considered for future research and classroom use.

This study adhered to the recommendations of Fernández Galeote and Hamari (2021), by focusing its analysis on individual games rather than the developers' intent, regardless of whether the games were designed for entertainment or as serious games. Interestingly, two of the top three recommended games, *Gibbon: Beyond the Trees* (*Gibbon: Beyond the Trees*, 2022) and *Beecarbonize* (Beecarbonize, 2023) can be classified as serious games. However, the classification of *Alba: A Wildlife Adventure* (*Alba: A Wildlife Adventure*, 2020) is less straightforward (Lu et al., 2023). Nevertheless, since entertainment games can also serve educational purposes (Djaouti et al., 2011) this study does not place significant importance on the distinction between serious and entertainment games.

4.5. Implications, limitations and future work

This study demonstrates the value of player-written reviews, left as voluntarily provided digital traces, for analyzing key human domains that influence behavior. By applying natural language processing (NLP) to these reviews, we could conclude about how players engage with climate change-related content, showcasing how individuals interact with video games and digital platforms to express their thoughts and experiences. Additionally, the analysis and comparison of playtime and game sentiment/popularity allowed us to explore user engagement and feedback toward ecogames, which are critical for understanding the adoption and impact of technologies (Hussain et al., 2019). By focusing

on the representation of climate change in player-written reviews of ecogames, this research also highlights the potential of AI-driven methods like NLP to address significant societal issues and advance our understanding of the intersection between human behavior and technology.

While this study aims to innovate in preselecting video games using NLP on player-generated reviews, there are limitations. As the data represents text generated by humans, its unstructured nature means that reviews can be noisy and even LLMs may struggle to extract meaningful information in some cases. Challenges such as gaming slang, satire, ASCII art, in-side jokes, and hyperbole further complicate the analysis. Additionally, many reviews are too short for meaningful text analysis; in our sample, 27.4 % of reviews consisted of fewer than five words and had to be excluded. Moreover, the definition of 'ecogame' is ambiguous and lacks precision, complicating the distinction from non-environmental video games and thus adding complexity to the identification process of representative ecogames. Notably, this challenge is not unique to ecogames; even more established genres such as action video games, which have undergone more thorough investigation (Bediou et al., 2023), grapple with the need for more explicit categorization when studying their effects on the human psyche (Dale et al., 2020). At the same time, this challenge stresses the potential of our NLP-based approach to study the effects of video games. Furthermore, the dataset used in this study was exclusively derived from player-written reviews of games available on Steam, thereby excluding mobile games, console games, and most serious games specifically designed for educational or research purposes. As a result, the presented NLP approach is most effective at detecting commercially available ecogames, limiting its generalizability to games developed by small independent creators, those originating from game jams, or certain serious games, which are often not commercially available.

Future work can refine the NLP methods utilized in this study. Further pretraining and/or fine-tuning ClimateBERT on Steam reviews may increase the model's familiarity with specific gaming language, potentially enhancing classification accuracy. Additionally, future research could focus on constructing and validating extensive climate change and environment-related dictionaries, addressing the omissions and inconsistencies that may arise from the LLM-based extension used in this study. Additionally, the dictionary-based approach might be enhanced by incorporating related terms using embeddings like Word2Vec (Mikolov et al., 2013). Moreover, the methodological approach taken in this study can be extended beyond environmental topics to identify games for education on economy, diversity, equality, and violence using zero-shot or few-shot classification approaches (Barrera Yañez et al., 2025; Ninaus et al., 2025). Another important avenue for future research is to evaluate the effectiveness of the games identified by the NLP approach as tools for environmental education and climate change communication. Finally, future work could aim to scale the NLP analysis to cover the entirety of Steam's game library, further broadening the scope and applicability of this approach. By doing so, we anticipate uncovering new, popular, and promising ecogames across various genres. This could contribute to the creation of a more comprehensive shortlist of ecogames, including details such as their popularities, genres, playtimes, and age ratings. While it remains uncertain how many ecogames can be identified by analyzing the entire Steam catalogue, we are confident that some relevant games will emerge. However, we acknowledge that well-liked, adequate ecogames may still be rare within the broader landscape of video games. Finally, this study lays the groundwork for developing an easy-to-use application that would allow educators to preselect and identify promising (eco-) games by specifying criteria such as playtime, genre, popularity, and age rating.

5. Conclusion

This study puts forward a natural language processing (NLP)

Table 9

Top three recommended ecogames based on NLP metrics and general game metrics.

Title	Popularity	Average Playtime	Dictionary analysis	Climate detection classification	Age rating
Alba: A Wildlife Adventure	98.2 %	4,65h	#16	#13	3
Beecarbonize	91.6 %	3,67h	#5	#5	3
Gibbon: Beyond the Trees	91.2 %	3,23h	#13	#16	12

Note: Rankings for the dictionary analyses and climate change detection classification are based on a total of 45 possible ranks, indicated by the hashtag '#'. Age ratings correspond to PEGI ratings.

methodology for identifying and preselecting ecogames that feature varying levels of climate change-related content. This approach offers a rapid, scalable, and data-driven method for preselecting games, benefiting both educators and researchers. By combining climate change detection classification with dictionary-based analysis, applied to over 186,000 Steam reviews, the methodology effectively distinguished ecogames from action/adventure games and highlighted those best suited for further research or educational use. The ability to differentiate ecogames from action/adventure games reinforces the practical utility and validity of ecogames as a thematic category, distinct from established genres like action/adventure, which may also include scattered ecogames. As such, the ecogame category serves as a valuable reference point for educators and researchers, independent of formal genre classifications. The preselection of games is crucial, as different ecogames did not equally feature content on environmental topics. Additionally, the methodology successfully identified novel candidate ecogames among randomly selected action/adventure titles, demonstrating its capability to uncover previously overlooked games. A key advantage of using NLP methods is their ability to analyze large volumes of qualitative data (i.e., player-written game reviews) quickly and at scale, a task that would be impractical to perform manually. By integrating NLP metrics with other game metrics, such as playtime and game popularity, we identified *Beecarbonize* (Beecarbonize, 2023), *Alba: A Wildlife Adventure* (Alba: A Wildlife Adventure, 2020) and *Gibbon: Beyond the Trees* (Gibbon: Beyond the Trees, 2022) as strong candidates for further empirical research and educational integration. These games are well-received, offer relatively short gameplay experiences, and engage players with environmental topics.

In summary, the proposed methodology has the potential to benefit researchers and educators by structuring the game selection process for classroom and laboratory use while providing a data-driven analysis of climate change-related content in ecogame reviews. As a result, even non-experts can identify promising and topical titles while keeping pace with the rapid developments in the video gaming industry. However, careful game selection remains essential. While a broader range of video games may be classified as ecogames based on the presence of climate change-related content, only a limited number effectively combine substantial climate change-related themes with widespread appeal and suitability for educational purposes.

CRedit authorship contribution statement

Moritz Edlinger: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Stefan E. Huber:** Writing – review & editing, Visualization, Formal analysis. **Manuel Ninaus:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this article, the authors used ChatGPT-4o to correct grammar mistakes and improve the overall fluency and readability of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Marie Lisa Kogler for her valuable feedback and insights. The authors would also like to thank Nathalie Scheucher for her help in the coding process.

The authors acknowledge the financial support by the University of Graz.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbr.2025.100850>.

Data availability

The code used for this study is available on OSF: https://osf.io/6hnpq/overview?view_only=13cbdb9ac829429187bb296f94241dab. Individual review data will be made available on request for other researchers.

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